

Computer Vision - Ball Counting, Detection and Pool-Table Retrieval

Francisco Silva, 202109000 José Guilherme Santos, 202208081 Mateus Cabral, 202511402
Victor Rodriguez, 202105114

Abstract—This report presents the methodology developed for Tasks 2 and 3 of the Computer Vision project. Task 2 estimates the number of visible pool balls using CNNs, comparing 17-way classification and bounded scalar regression across several architectures trained from ImageNet initialization and from scratch. Task 3 addresses ball detection, comparing YOLOv8n with DETR-ResNet50, and pool-table retrieval, where a query image is matched to the most similar table state. Retrieval uses HSV and ResNet18 baselines, followed by a table-focused method based on top-down normalization, CLIP embeddings, a count-difference penalty and a YOLO-based fallback when normalization fails.

I. INTRODUCTION

Building on the ball-detection and top-down transformation pipeline from Task 1, this stage moves from classical image processing to learning-based models. Task 2 asks for a CNN that predicts only the total number of visible balls, while Task 3 asks for individual ball detection and retrieval of the most similar table state from the training data. These problems are difficult because the dataset is small, the balls occupy a small image region, and the images vary in viewpoint, lighting, background and table visibility. Therefore, the main design choice was to reduce irrelevant visual variation: counting uses aspect-ratio-preserving resizing and automatic table cropping, while retrieval uses top-down normalization so that images are compared as table states rather than full camera views.

II. THEORETICAL BACKGROUND

A. Convolutional Neural Networks and Transfer Learning

CNNs combine convolutional filters, nonlinear activations and pooling to learn hierarchical visual features, from edges to object-level patterns [1]. Since this dataset contains only a few hundred images, transfer learning is important: ImageNet-pretrained backbones already contain useful low- and mid-level visual filters that can be fine-tuned for ball counting [2].

B. Object Detection: YOLO and DETR

YOLO treats detection as a single-stage prediction problem and enables fast inference [3]. YOLOv8 uses an anchor-free, decoupled detection head and is suitable for small-object detection with limited data [4]. DETR formulates detection as set prediction using a transformer encoder-decoder and bipartite matching loss [5]. This removes anchors and NMS, but generally requires more data and longer training, which is a disadvantage here.

C. Image Embeddings for Retrieval

Image retrieval represents each image by a feature vector and ranks candidates using a similarity measure such as cosine similarity. CLIP is trained with an image-text contrastive objective and produces embeddings that capture high-level semantic content [6]. In this project, CLIP is applied after top-view normalization to focus retrieval on the table state instead of background, camera viewpoint or broadcast overlays.

III. TASK 2: BALL COUNTING WITH CNNs

A. Dataset and Preprocessing

The dataset used in this project is the Roboflow Universe 8-Ball Pool object-detection dataset, published by BachelorThe-sis under a CC BY 4.0 license [7]. It contains 247 labeled images with five classes: Dot, Black, Cue, Solid and Striped. Ground-truth counts are obtained from the YOLO label files by counting only the four physical ball classes: Black, Cue, Solid and Striped. The Dot class is ignored because it is not a physical ball to be counted.

Two preprocessing steps were used. First, letterbox resizing preserves aspect ratio and pads the remaining area, avoiding ball deformation. Second, an HSV/table-mask procedure automatically crops the dominant table-cloth region, reducing background information and increasing the relative size of the balls. The same crop is applied during training, validation, testing and inference. Training also uses mild horizontal flips, color jitter and occasional Gaussian blur.

B. Counting Formulations

Two formulations were compared. In the classification formulation, the final layer outputs probabilities for counts from 0 to 16, allowing uncertainty over plausible counts. In the bounded regression formulation, called *logistic* in the code, the network outputs one scalar logit; a sigmoid maps it to $[0, 16]$, and the result is rounded:

$$\hat{y} = \text{round}(16 \sigma(z)). \quad (1)$$

This uses the ordered nature of counts, since an error of one ball is less severe than a large count error.

C. Architectures and Training

Twelve CNN backbones were trained: AlexNet, VGG16, ResNet18, ResNet34, ResNeXt50, ConvNeXt-Tiny, EfficientNet-B0, MobileNetV2, MobileNetV3-Large, DenseNet121, ShuffleNetV2 and SqueezeNet1.1. Each architecture was trained under both counting formulations

and with two initialization strategies: ImageNet-pretrained and random initialization. This produced 48 models in total.

Pretrained models used a smaller learning rate for the backbone and a larger learning rate for the newly initialized prediction head. Models were checkpointed according to validation MAE, and the test set was used only for final reporting. For readability, the result table is ordered by exact test accuracy, then within-one accuracy, MAE and RMSE.

TABLE I: Task 2 training configuration.

Setting	Value
Input size	512
Maximum count	16
Counted classes	Black, Cue, Solid, Striped
Ignored class	Dot
Resize	Letterbox
Table crop	Automatic HSV/table-mask crop
Epochs / batch size	60 / 4
Early stopping patience	12
Optimizer	AdamW
Classification loss	Cross-entropy with class weights
Bounded regression loss	Smooth L1
Pretrained backbone LR	10^{-5}
Pretrained head LR	10^{-4}

Because the ball-count distribution is imbalanced, class imbalance was handled in two ways. First, a WeightedRandomSampler rebalanced the training batches for all models. Second, for classification models, the cross-entropy loss was weighted by inverse class frequency. For bounded regression models, the sampler still helps, but class-weighted cross-entropy does not apply because the loss is Smooth L1.

D. Baselines and Evaluation

Three constant baselines were computed: rounded training-set mean, median and mode. Models were evaluated using MAE, RMSE, exact-count accuracy, within-one accuracy ($|\hat{y} - y| \leq 1$), mean signed error and R^2 . Exact accuracy is useful but unstable on a small test set, so MAE and within-one accuracy are also reported.

E. Results and Error Analysis

Table II reports the five best CNN counting variants on the test set, together with the constant mean baseline. This wording avoids the earlier misleading claim that the table shows the best variant per architecture.

TABLE II: Task 2 test-set ball-counting results: five best CNN variants and constant mean baseline.

Model	MAE	± 1 Acc.	Exact Acc.
VGG16, bound. reg., ImageNet	1.25	0.65	0.40
AlexNet, class., ImageNet	1.45	0.70	0.35
AlexNet, bound. reg., ImageNet	1.45	0.70	0.25
AlexNet, bound. reg., scratch	1.45	0.65	0.25
DenseNet121, bound. reg., scratch	1.45	0.75	0.10
Constant baseline, mean	3.15	0.30	0.10

The CNNs substantially outperform the baseline, reducing MAE from 3.15 to approximately 1.25–1.45. Most errors are near misses, such as predicting 13 instead of 14. The hardest

cases contain clustered or occluded balls, very small balls due to perspective, or failed crops that leave distracting background information.

IV. TASK 3: BALL DETECTION

A. Dataset Split

For Task 3, the original 247 images were reorganized into a dedicated YOLO folder structure with 197 training images, 24 validation images and 26 test images. Detection uses only the four physical ball classes: Black, Cue, Solid and Striped. The Dot class is removed so that detection and count-level evaluation correspond to actual balls.

B. YOLOv8n

YOLOv8n was used as the main detector because it is lightweight and well suited for a small dataset with small objects. It was trained at a fixed input resolution with early stopping based on validation performance, and inference was restricted to the four physical classes. YOLOv8s was considered as an extension but was not included in the final comparison.

C. DETR-ResNet50

DETR-ResNet50 pretrained on COCO was adapted by replacing its classification head with a four-class head. The model was fine-tuned with AdamW using a small learning rate, keeping the checkpoint with the lowest validation loss. Both detectors were evaluated on the same no-Dot test set.

D. Results

Table III reports the test-set comparison. YOLOv8n clearly outperformed DETR-ResNet50 on both localization and count-level metrics.

TABLE III: Detection comparison on the four physical-ball classes.

Model	mAP50	mAP50:95	Rec.	MAE	Exact	± 1
YOLOv8n	0.9200	0.7654	0.7950	0.5385	0.6923	0.8846
DETR-R50	0.1573	0.1104	0.2466	9.3462	0.0000	0.0000

DETR was highly sensitive to the confidence threshold. At 0.25 it over-predicted, producing about 20 detections per image for images containing about 11 balls; at 0.50 it under-predicted and produced almost no detections. This indicates poor calibration and insufficient adaptation to small, densely clustered objects.



Fig. 1: YOLOv8n detection examples. The left image shows a zero-count-error prediction, where the predicted number of balls matches the ground truth. The right image shows a higher-error prediction, illustrating typical failure modes such as missed balls, occlusion and false positives.

V. TASK 3: POOL-TABLE RETRIEVAL

The retrieval task asks for the most similar training table given a query image. Three retrieval strategies were implemented, plus a fallback for failed top-view transformations.

A. HSV Histogram Baseline

Each image is converted to HSV and represented by a normalized 3D color histogram. Candidates are ranked by cosine similarity. This baseline is simple and explainable, but it mainly compares global color distribution and is easily misled by lighting, background, broadcast overlays and camera angle.

B. CNN Embedding Baseline

A pretrained ResNet18 with its classification head removed is used as a feature extractor, again with cosine similarity. This captures more visual structure than HSV histograms, but still focuses on the whole scene rather than specifically on the ball arrangement.

C. Top-View Normalization and CLIP Retrieval

The main method first reuses the Task 1 top-down transformation, because the goal is to compare table states rather than raw camera views. This reduces the influence of viewpoint and background. Since the perspective transform can fail, each top view is checked using black ratio, cloth-region size, plausible aspect ratio and whether the output is merely a resized copy of the original. Only accepted top views are kept in the retrieval database.

CLIP ViT-B/32 embeds the accepted top views. Retrieval combines CLIP cosine similarity with a count-difference penalty:

$$\text{score}(q, i) = \text{sim}_{CLIP}(q, i) - \alpha |c_q - c_i|, \quad (2)$$

where c_q is the query count and c_i is the candidate count. This downranks visually similar images with very different game states.

D. Ball-Layout Fallback

If the top-view quality check rejects a query image, the system switches to a ball-layout fallback. YOLOv8n detects balls in the original image, an HSV cloth mask estimates the table boundary, and detected centers are normalized to table-relative coordinates. The centers are binned into a fixed-size 32-dimensional occupancy descriptor. Cosine similarity between descriptors, combined with the count penalty, enables retrieval when perspective normalization is unavailable.

E. Results

Retrieval is evaluated qualitatively, with absolute count difference used as a weak proxy to identify clearly wrong matches. The HSV and ResNet18 baselines sometimes retrieve images with similar color or background but a different ball layout. Top-view CLIP produces more meaningful matches because it compares normalized table states and penalizes count mismatches. The fallback is useful when the perspective

TABLE IV: Qualitative retrieval summary.

Method	Strength	Main weakness
HSV histogram	Simple color baseline	Sensitive to lighting/background
ResNet18 embedding	Captures global scene structure	Viewpoint- and background-dependent
Top-view CLIP	Best table-focused comparison	Depends on perspective transform
Ball-layout fallback	Used when top view is rejected	Depends on YOLO and table mask

transform fails, although it depends on YOLO detections and the cloth mask.

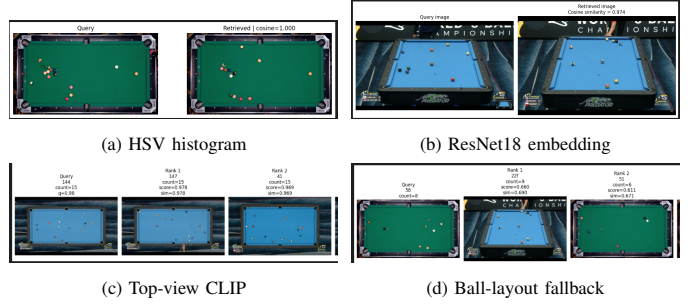


Fig. 2: Qualitative comparison of the retrieval methods. HSV histogram and ResNet18 embedding retrieval mainly compare global appearance and can be affected by lighting, background and viewpoint. Top-view CLIP compares normalized table states, while the ball-layout fallback uses YOLO-detected ball positions when the top-view transformation is unreliable.

VI. CONCLUSION

The best ball-counting results were obtained by CNNs trained with aspect-ratio-preserving resizing and table-focused preprocessing, confirming that input normalization matters when balls occupy a small image region. Recasting the problem as either 17-way classification or bounded regression enabled a systematic comparison of twelve CNN backbones under pretrained and from-scratch initialization.

For detection, YOLOv8n clearly outperformed DETR-ResNet50 on both localization and count-level metrics, achieving high mAP and low count MAE on the no-Dot test set. DETR struggled because the dataset is small and the objects are small and clustered. For retrieval, global HSV and CNN embeddings were useful baselines but too sensitive to background, lighting, overlays and camera angle. The strongest strategy used Task 1 top-down normalization, CLIP embeddings and a count penalty, with a YOLO-based ball-layout fallback when perspective normalization was unreliable.

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